DEECAMP

Cell Lore × Cell Eclipse

Every Cell Has a Story

scATAC-seq | Multiome | Foundation Model

Self-supervised representation learning for sparse regulatory signals, powered by sc-agent data curation.

For what we already know.

For what we can imagine.

Cute_niuma

卢绍荣
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Genomics Models
Transcript Regulation
Multi-Omics

符源
Yuan Fu

Surface Protein
Proximity Omics
Multi-Omics

谷峰
Feng Gu

Transcript Regulation
Multi-Omics

覃思颖
Siying Qin

Protein Design
Multi-Omics

隋烽瑞
Fengrui Sui

CITE-seq
Multi-Omics

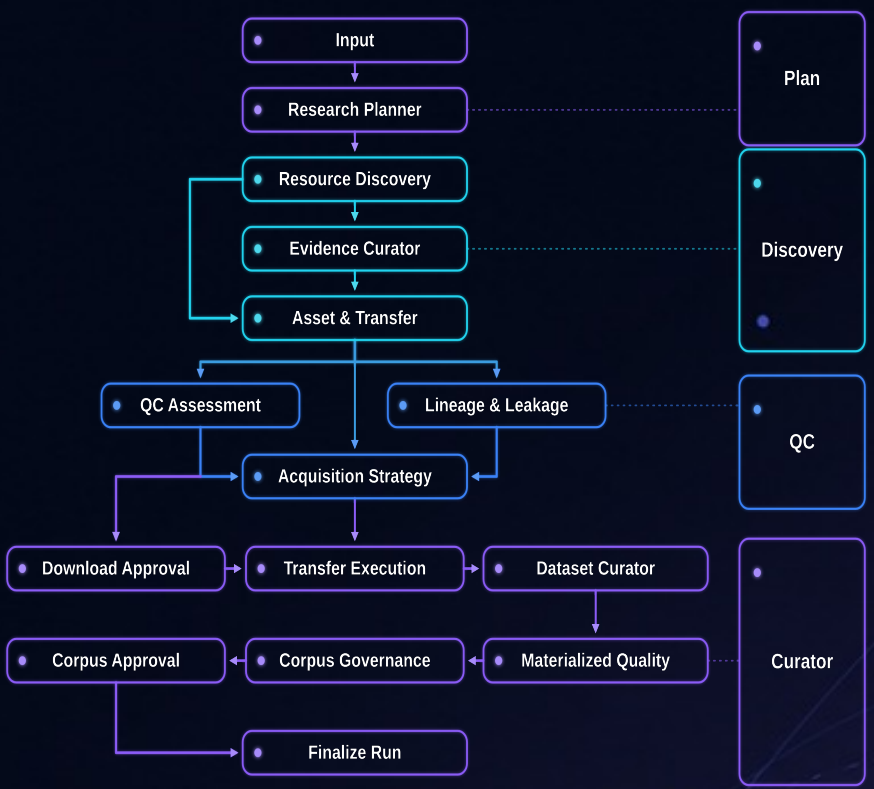
CONTRIBUTORS

DEECAMP

Cute_niuma

HuPMO
powered by CellLore

Corpus Design Agent for Single-Cell Foundation Models



113
Independent Studies

1,028
RNA–ATAC Paired Data Units

8,975,454
Landed Cells on Disk

60
Tissues

SAME-CELL PAIRING

11.7×
VS SCMMO-ATLAS



scMMO-atlas	768,459
scMMDB	640,192

Published paired-subset counts are shown for comparison; general atlas totals are not used as the denominator.

01 RNA ↔ ATAC
Cell-Resolved Pairing
Every RNA profile remains explicitly linked to its ATAC measurement.

02 COUNTS ↔ FRAGMENTS
Raw Signals Retained
Reusable RNA counts and indexed ATAC fragments are preserved.

03 SOURCE ↔ WORKFLOW
Uniform Reprocessing
Heterogeneous deposits converge on one reproducible workflow.

04 TISSUE ↔ DISEASE
Biology-Aware Curation
Donor, tissue, disease and treatment are harmonized as first-class data.

05 UNIT ↔ SCHEMA
Standardized Data Model
Every validated paired unit follows the same archival contract.

Yuan Fu & Siying Qin

HuPMO

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Human biology across tissues, technologies and states

Specific tissues

Cerebellar cortex	901,536
Putamen	774,258
Middle temporal gyrus	744,610
Posterior cingulate gyrus	710,937
Substantia nigra	693,898
Retina	389,821
Prostate	355,458
Colon	316,602

Organ systems

Nervous & sensory	4,664,347
Hematopoietic & immune	879,262
Reproductive & breast	688,621
Digestive	621,975
Developmental models	542,758
Musculoskeletal	383,480
Respiratory	310,460
Cardiovascular	273,086

Assay families

10x Multiome 868	SHARE-seq 70	DOGMA-seq 53	Unspecified RNA + ATAC multiome 24
CAT-ATAC 5	TEA-seq 5	NEAT-seq 2	SNARE-seq2 1



Disease status resolved

8,798,553 cells



Treatment condition resolved

8,423,255 cells



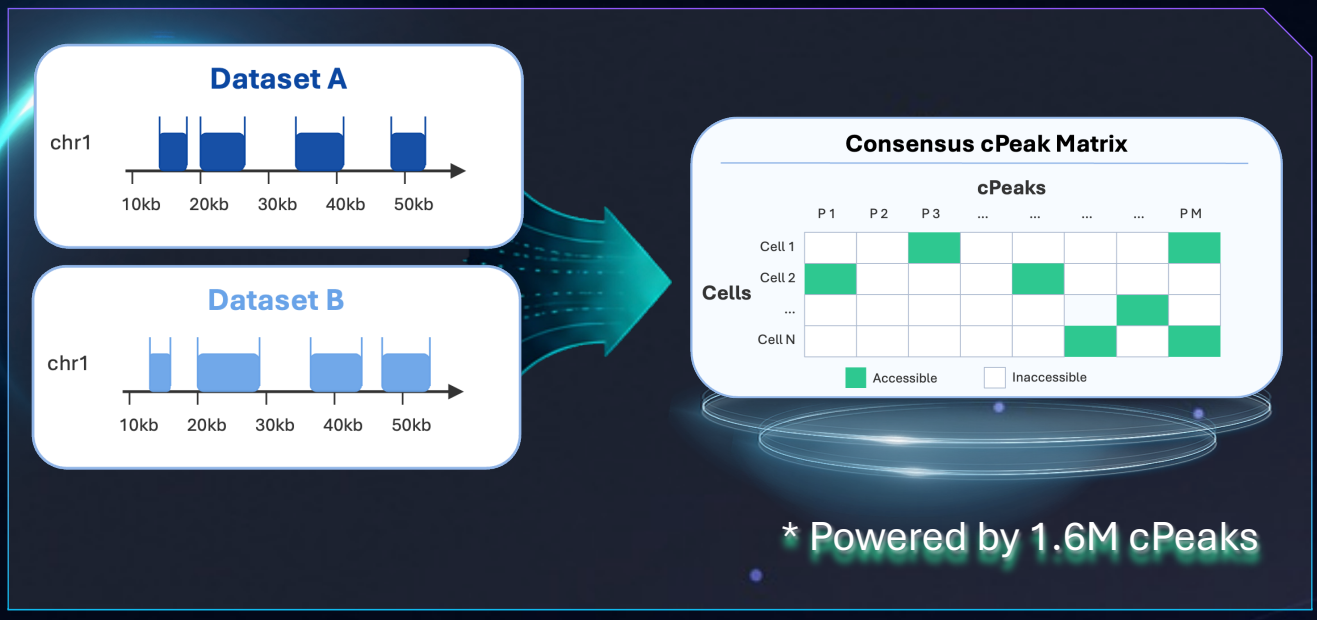
Complete biological context

8,036,697 cells

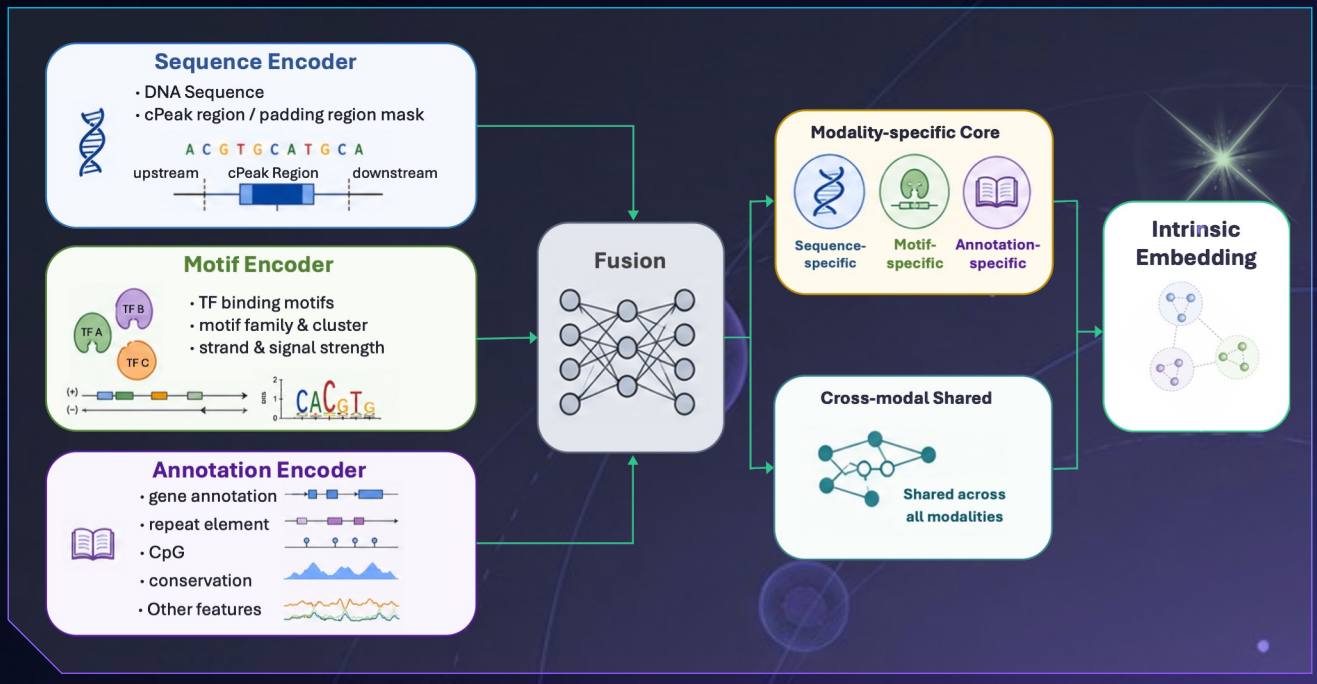
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Unified ATAC feature axes



Intrinsic Peak Tokenizer



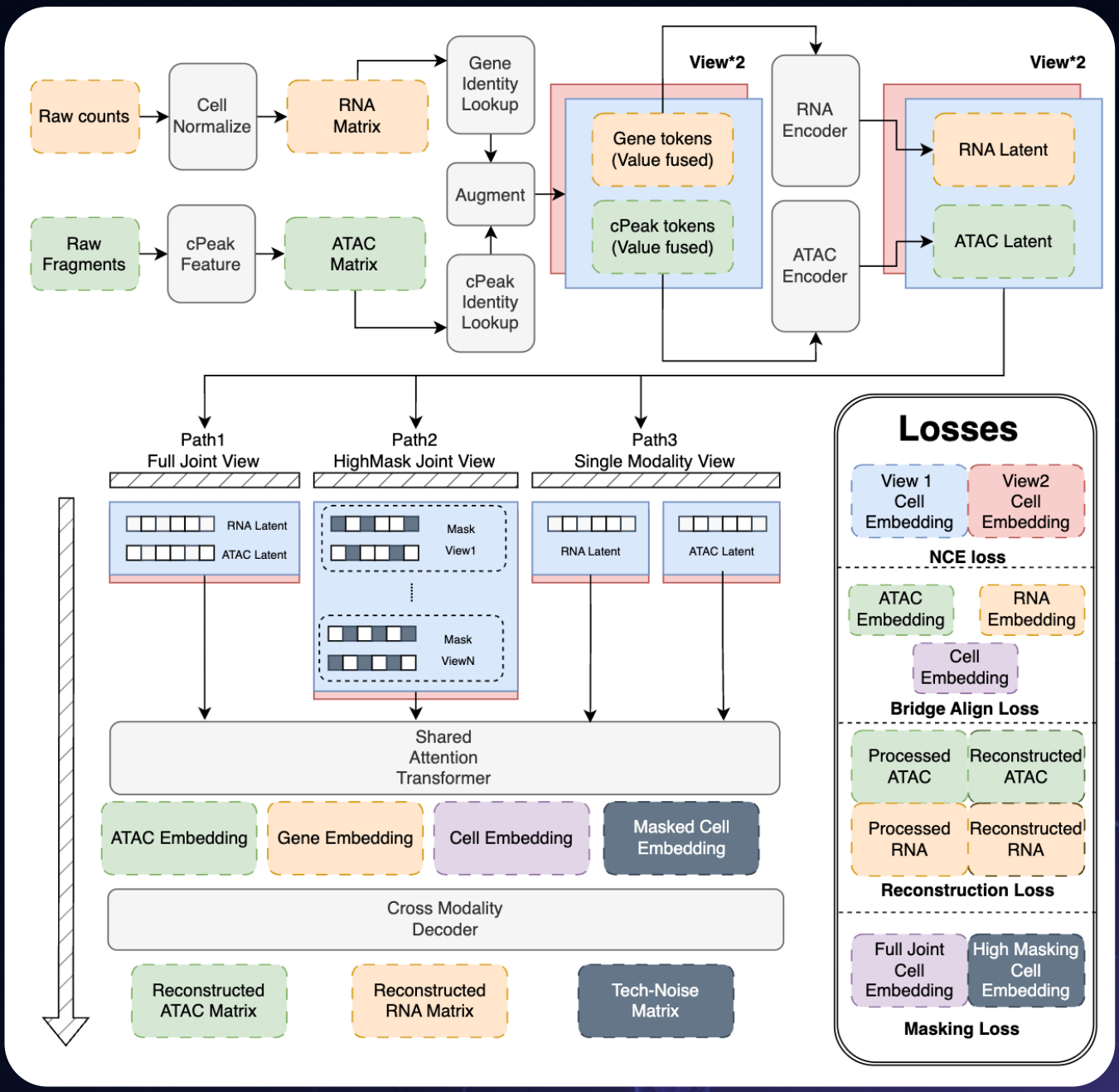
Regulatory Knowledge Captured

Biologically Informed Peak Representations

Feng Gu

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Cell Eclipse Architecture



* InfMasking Theory
* Contrastive Learning
* Cell-Reconstruct
* Modality Alignment
* Tn5-Bias Augment

Shaorong Lu

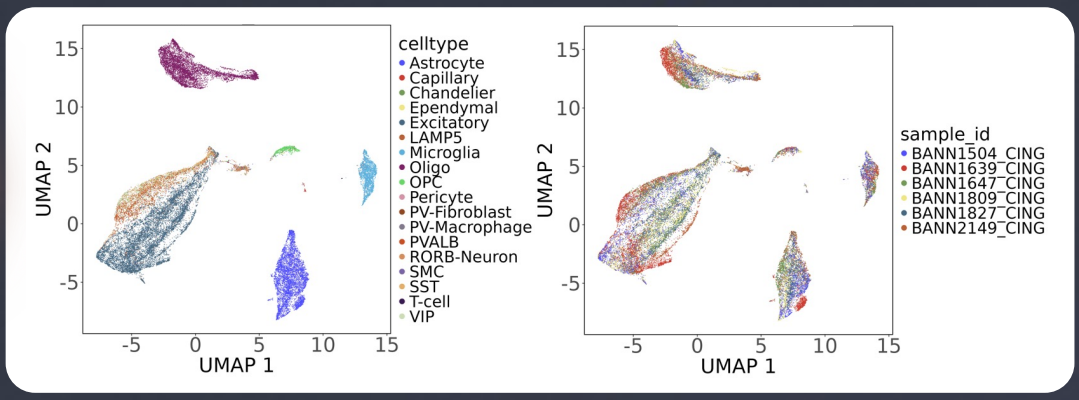
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Benchmark

Test-dataset UMAP

Embedding

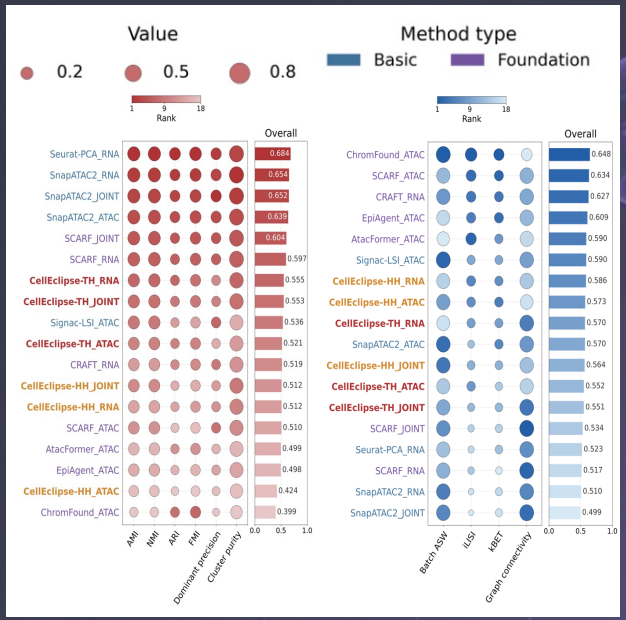
Known Label



Zero-Shot Cell Clustering

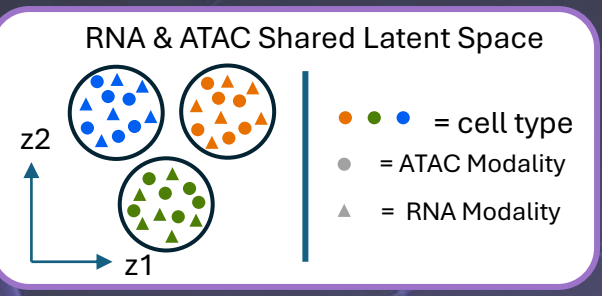
Frozen Cell Eclipse Encoder

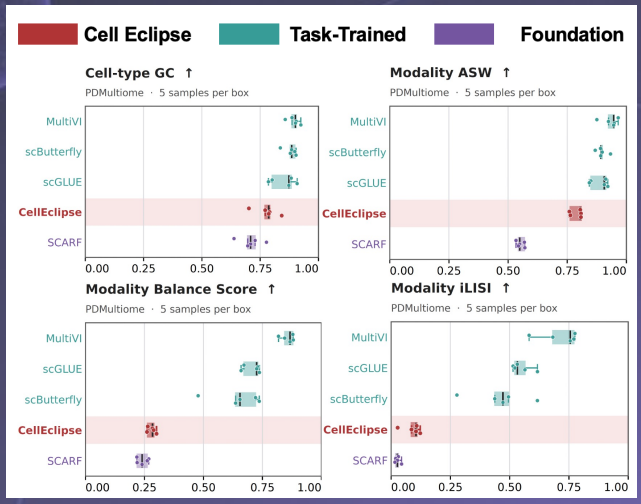
K-Means Clustering



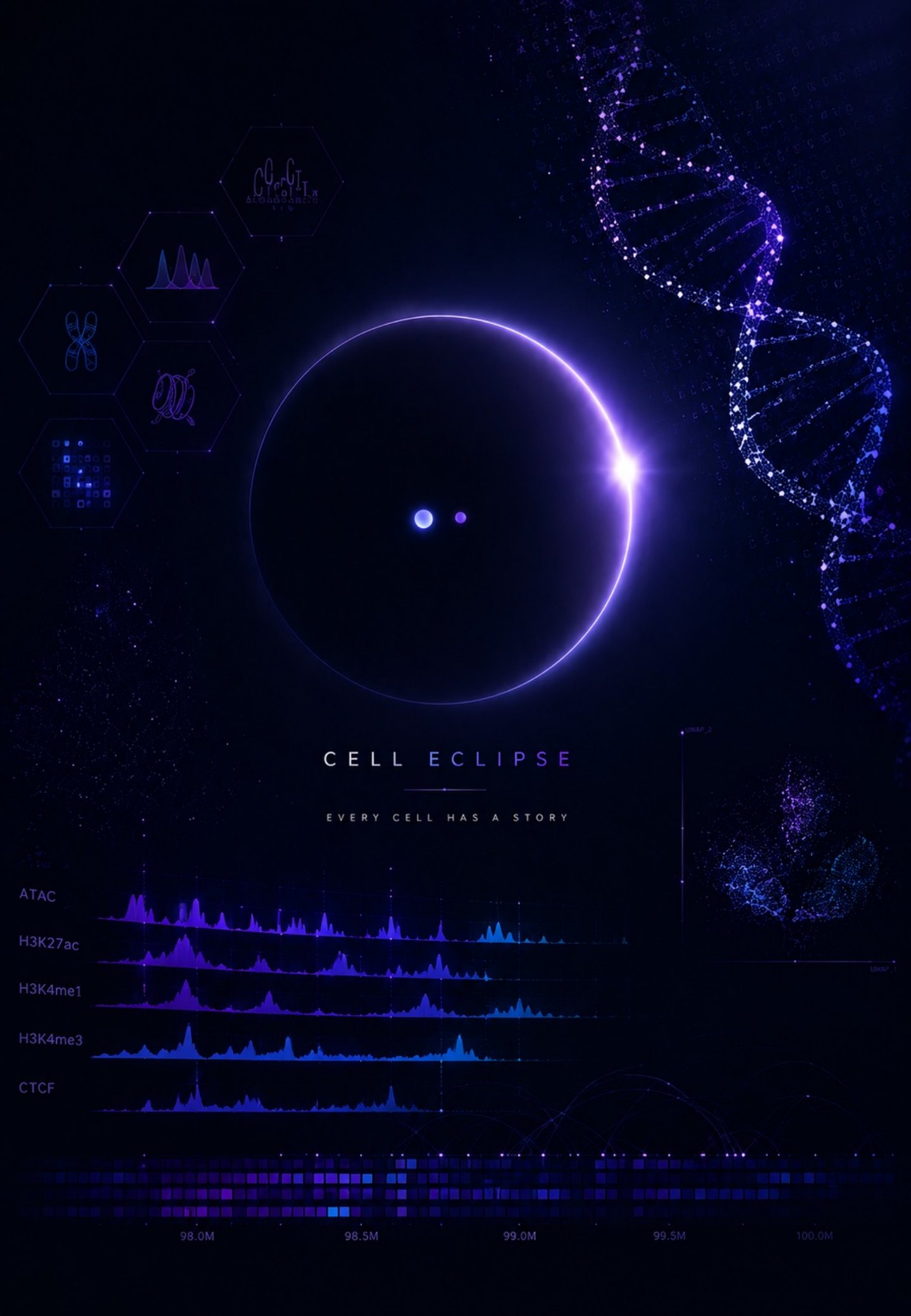
Joint Representation

Frozen Cell Eclipse Encoder





Fengrui Sui


CELL ECLIPSE
EVERY CELL HAS A STORY
ATAC
H3K27ac
H3K4me1
H3K4me3
CTCF
98.0M
98.5M
99.0M
99.5M
100.0M